SOLUTIONS: DISCRETE EVENT SYSTEMS
MASTER IN CONTROL

1. Exercise

a) The storage facility can keep at most three boxes of type 1, or two boxes of type 2. Alternatively it could also fit one box of type 1 and one of type 2. Accordingly the automaton G is shown in Fig. 1.1. [4]
(2 marks for correct events and states; 2 marks for correct transition diagram).

b) Sometimes events of type $\tilde{a}_1$ occur. These trigger exactly the same transition as an event of type a_2 as they corresponds to arrivals of boxes of size 3. Hence the automaton $\tilde{G}$ is shown in Fig. 1.2. [3]

c) The parallel composition G_P is shown in Fig. 1.3. [4]

(1 mark for correct identification of events and states; 3 marks for correct transition diagram)

d) The non-deterministic automaton G_N is shown in Fig. 1.4. In order to design the diagnoser we remark that there are no ε transitions in G_N, hence there is no need to compute ε-reaches when building the next state visited by any initial set of states. For the sake of simplicity we collapsed all states including only Y states within a single state Y. The corresponding automaton is shown in Fig. 1.5. [6]
(2 marks for correct non deterministic automaton, 1 mark for correct states and events of the observer automaton, 3 marks for correct transition diagram)

e) The string $a_1a_1a_1d_1$ is classified as an N string, since 3 consecutive a_1 events (which indeed saturate the store capacity) are only possible if these are genuine a_1s. Finally the event d_1 is never misclassified. The string $a_2a_1d_2d_2$ is classified as a Y string since two d_2 events have occurred but only one a_2 has preceded them. Hence the only possibility is that the a_1 event was a case of misclassification. Finally a_2a_1 is classified as a YN string, since both possibilities are open. In particular there would be enough capacity in store to have two boxes of size 3 and of course also one of size 2 and one of size 3. [3]

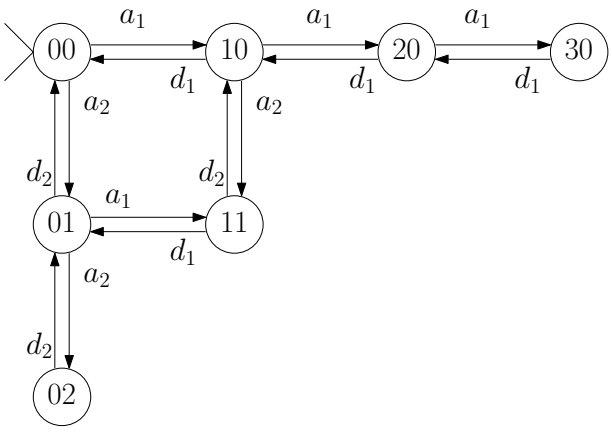


Figure 1.1 The automaton G modelling the storage

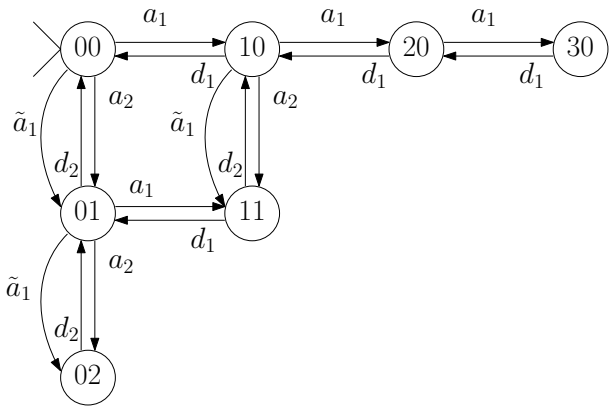


Figure 1.2 The automaton $\tilde{G}$ modelling storage with misclassification events

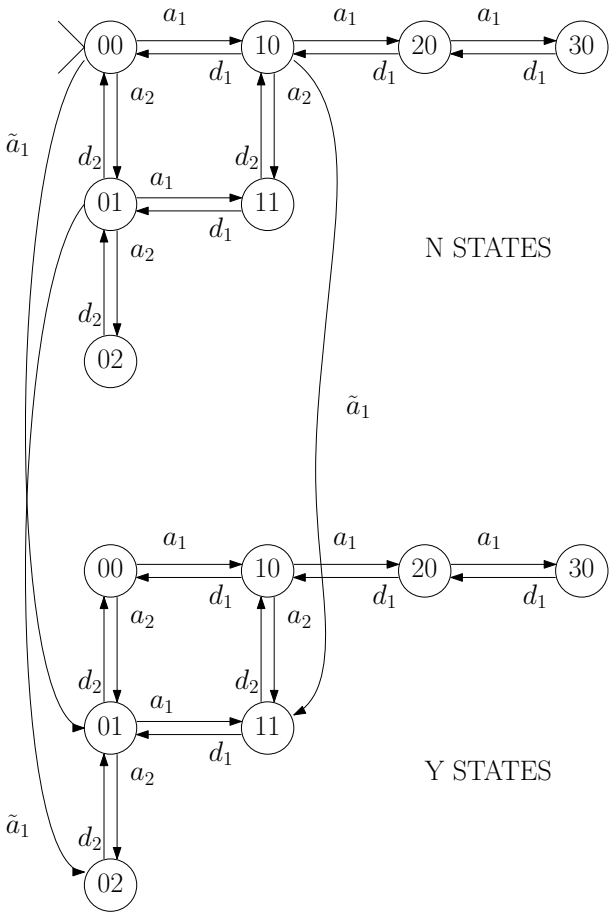


Figure 1.3 The parallel composition G_P

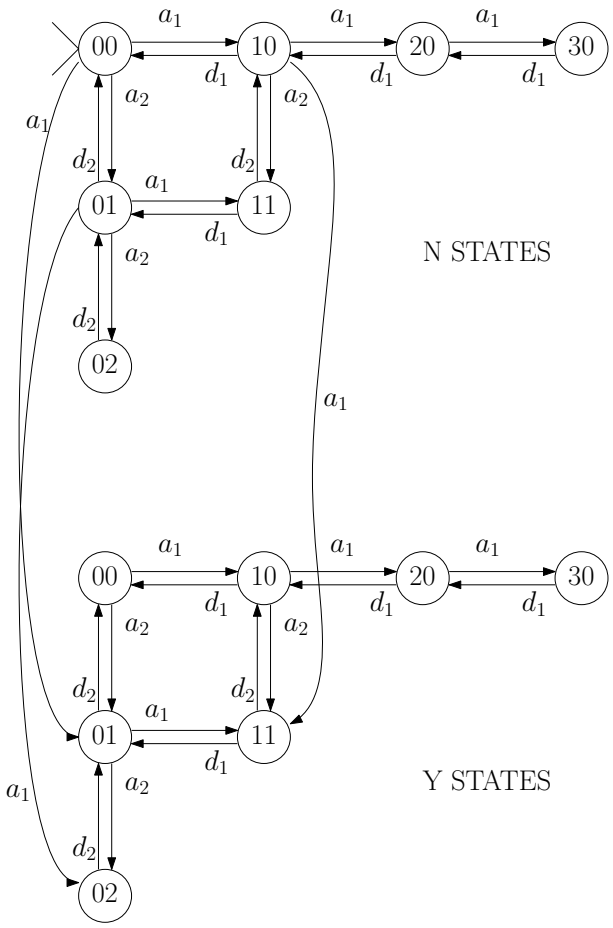


Figure 1.4 The non deterministic automaton G_N

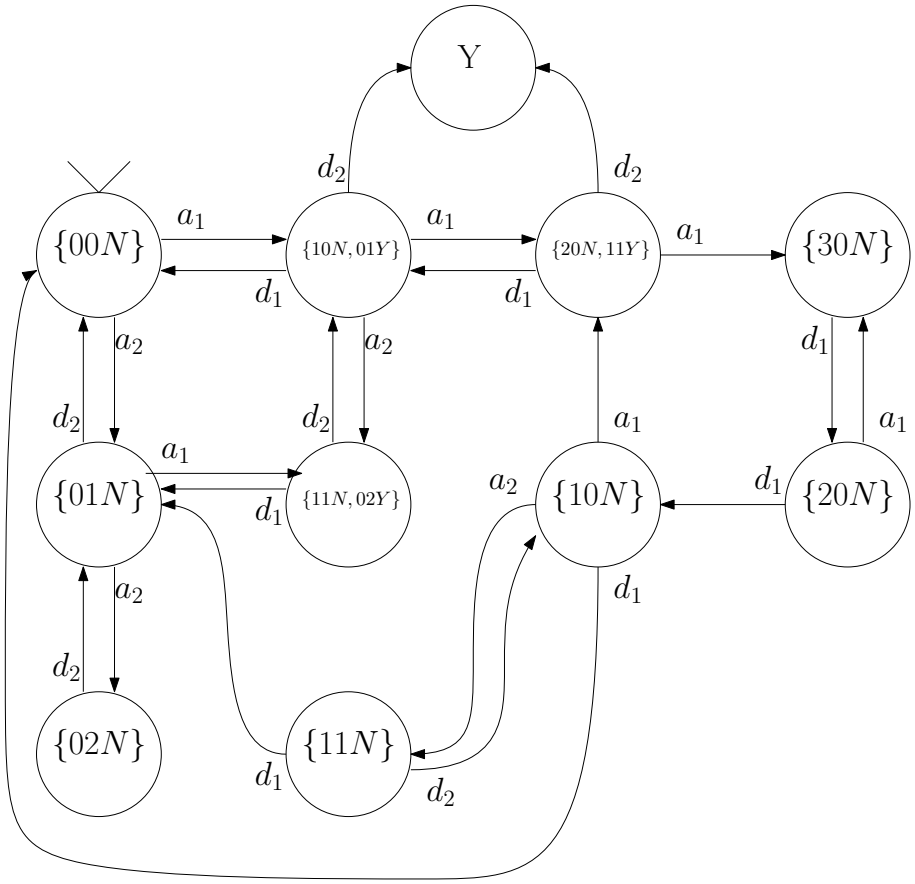


Figure 1.5 The diagnoser

2. Exercise

a) The automata G_1 and G_2 are shown in Fig. 2.1. [3]

b) The Robot is modeled with the Automaton G_R shown in Fig. 2.2. [3]

c) The parallel composition is shown in Fig. 2.3. [4]
(1 mark for correct identificaiton of events and states; 3 marks for correct transition diagram).

d) A cycle of the specification should allow occurrence of u_1 followed by u_2 or viceversa. We let the initial state be F to denote that assembling operations are Finished. We then allow event u_1 or u_2 to occur, leading respectively to states A and B, where u_2 or u_1 are supposed to bring us back to the original state F. The specification automaton G_{spec} is shown in Fig. 2.4. [3]

e) In order to design a non-blocking supervisor it is enough to compute the parallel composition $G||G_{spec}$ and do the Trim of the resulting automaton. All events are controllable, hence no controllability issues will arise. [3]

f) The parallel composition is shown in Fig. 2.5, while Trim($G||G_{spec}$) is in the following Fig. 2.6. [4]
(1 mark for correct identification of events and states, 3 marks for correct transition diagram).

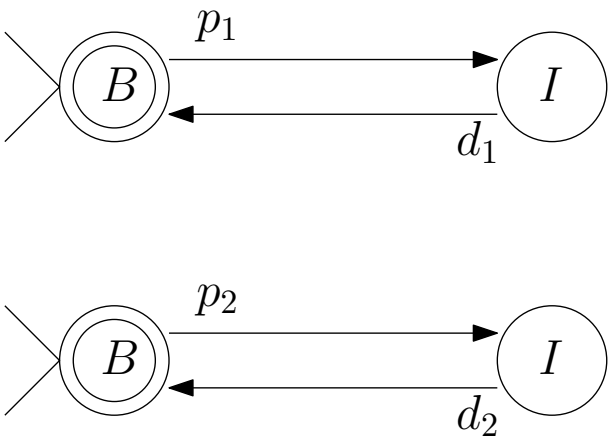


Figure 2.1 The automata G_1 and G_2

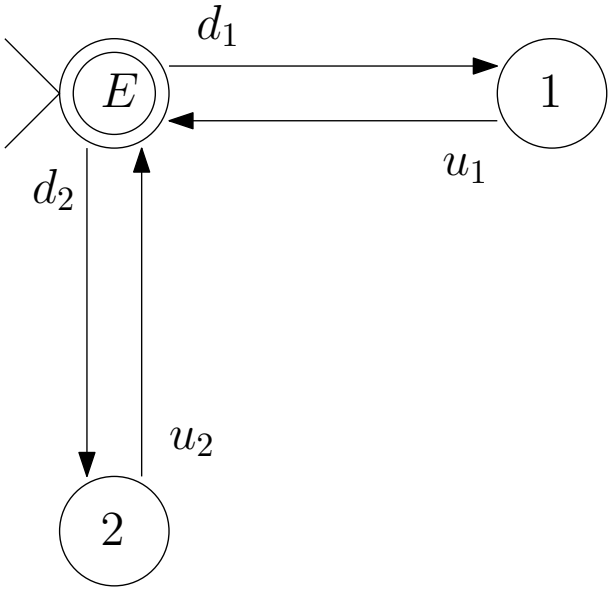


Figure 2.2 The automaton G_R

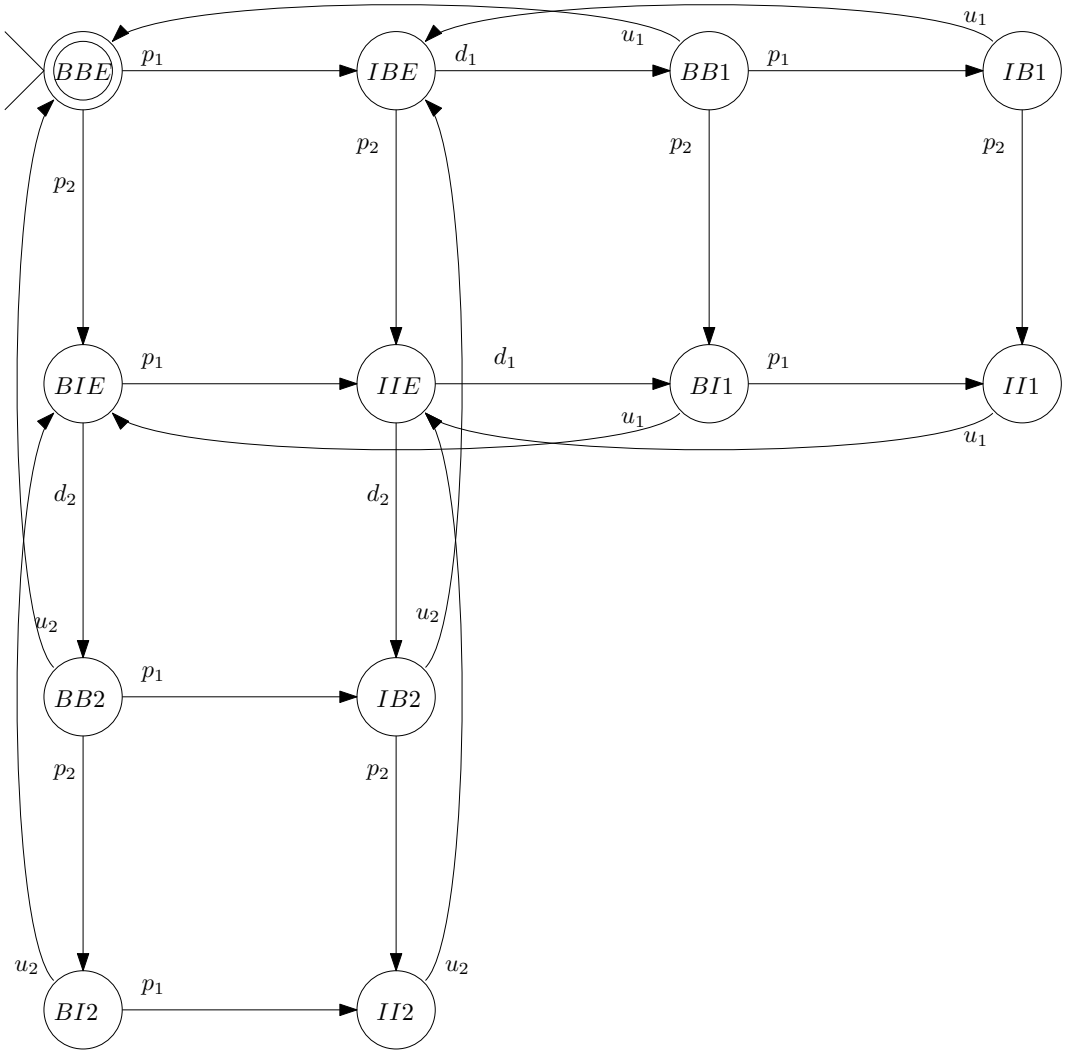


Figure 2.3 The automaton *G*

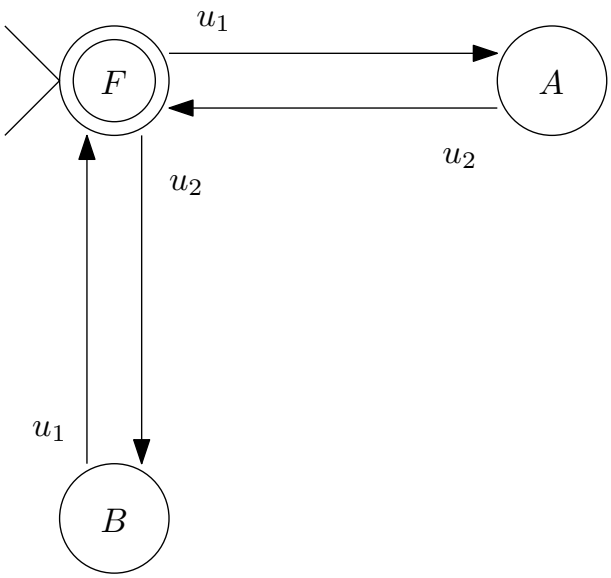


Figure 2.4 The automaton G_{spec}

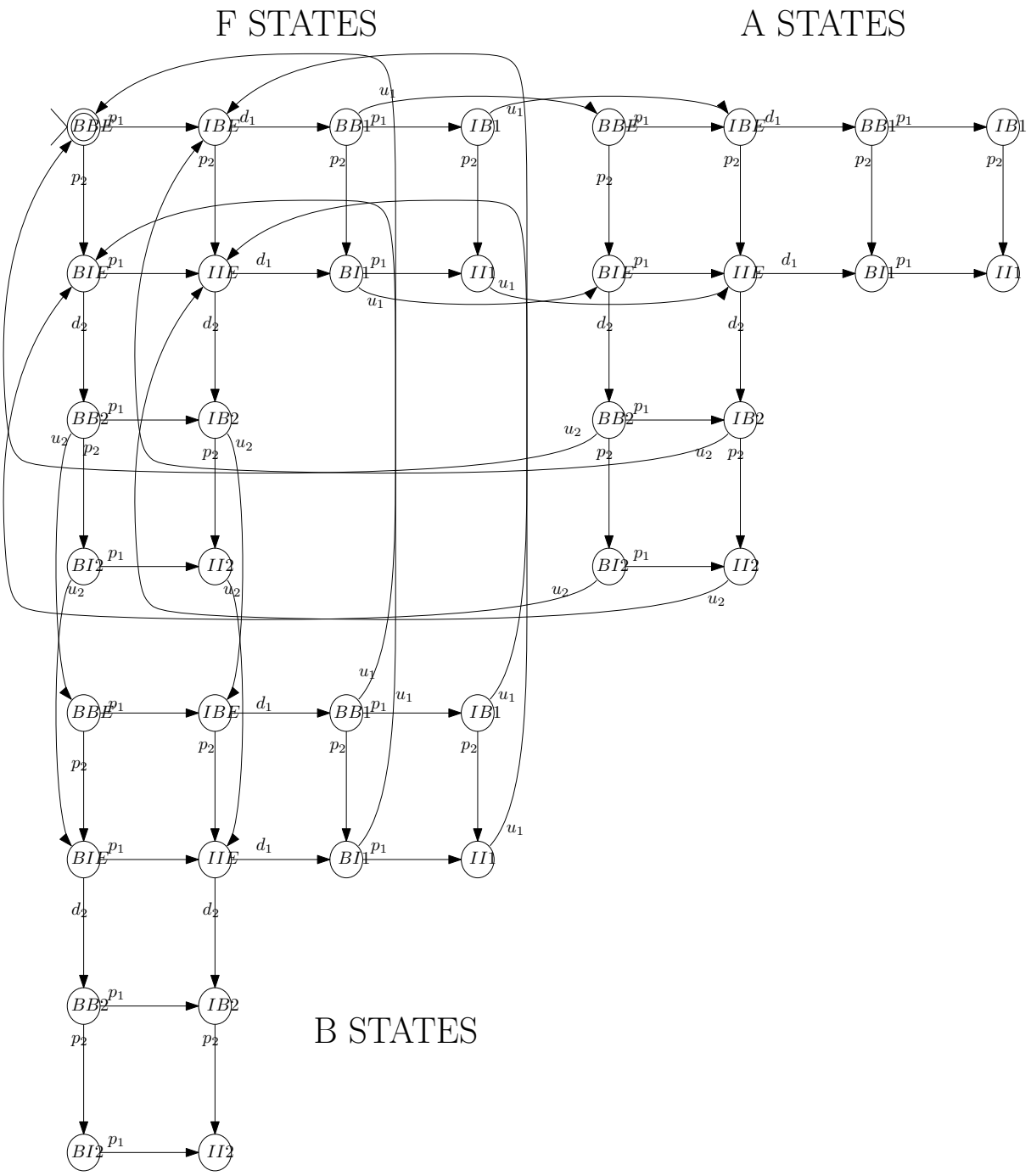


Figure 2.5 The automaton $G||G_{spec}$

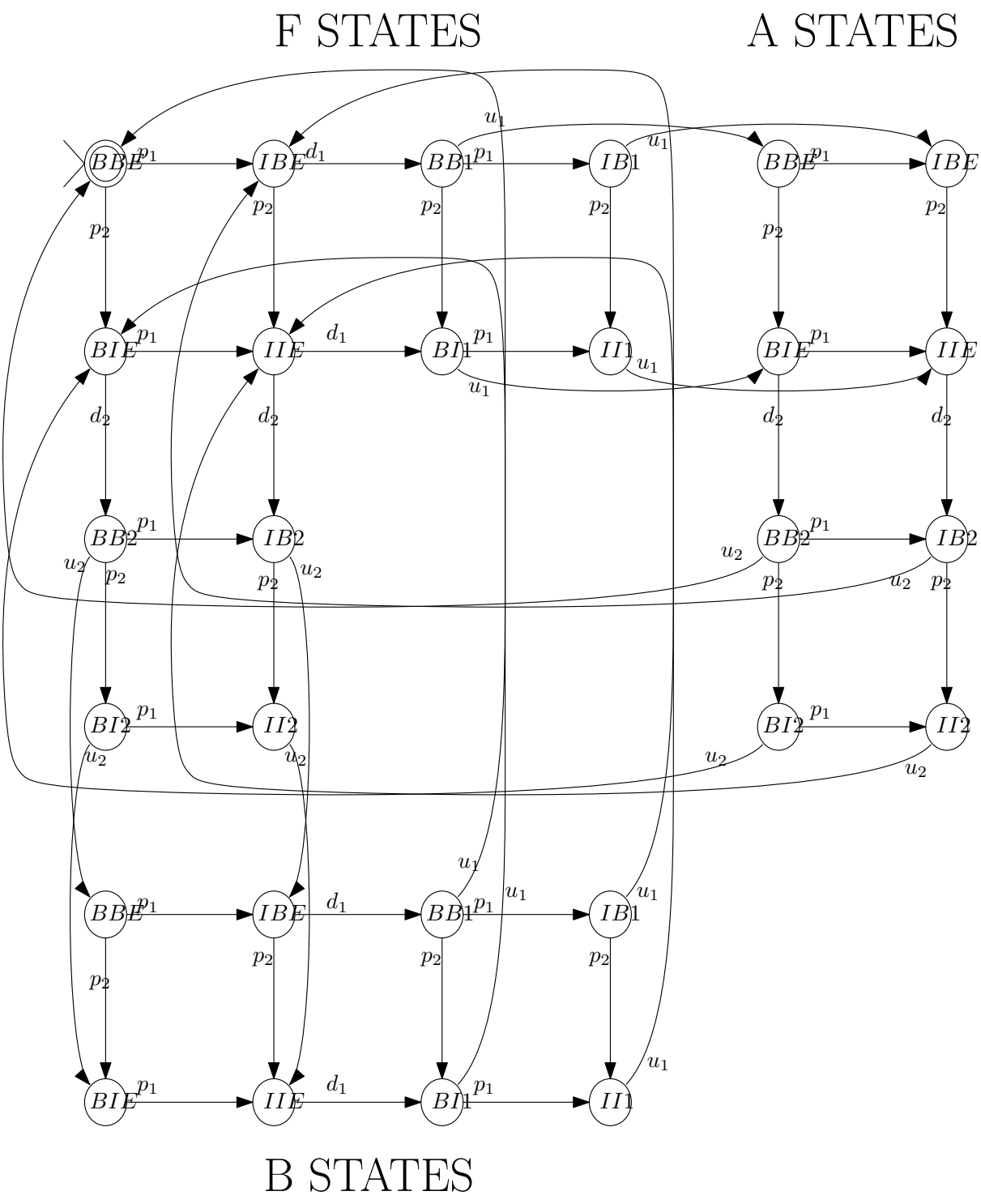


Figure 2.6 The automaton Trim($G||G_{spec}$)

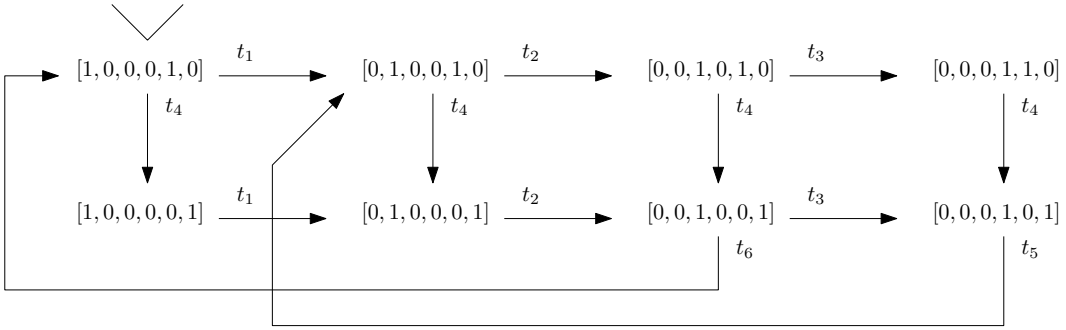


Figure 3.1 Transition diagram associated to $\mathscr{R}(M_0)$

3. Exercise

a) Ordering the species in alphabetical order yields the following incidence matrix:

$$C = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 1 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 1 & -1 & -1 \end{bmatrix}.$$

[4]

(2 marks for correct Pre matrix and 2 marks for correct Post matrix).

b) The Petri Net N admits two independent minimal P-semiflows, $v_1 = [1,1,1,1,0,0]$ and $v_2 = [0,0,0,0,1,1]$. These correspond to conservation laws in the concentration of species. In particular $A+B+C+D = const.$ and $E+F = const.$ [4]
(2 marks for correct P-semiflows and 2 marks for correct interpretation).

c) Notice that $v_1 + v_2 = [1,1,1,1,1,1]$ is a P-semiflow and its support is $\{A,B,C,D,E,F\}$. Hence N is structurally bounded and structurally conservative. [2]

d) The Petri Net N admits 2 independent T-semiflows, $[1,1,0,1,0,1]'$ and $[0,1,1,1,1,0]'$. They identify sequence of reactions that leave concentrations unchanged. For instance $t_1t_2t_4t_6$, (provided there are enough molecules/tokens to carry out such reactions). [4]
(2 marks for correct T-semiflows and 2 for correct interpretation).

e) The reachable set has cardinality 8 and the corresponding transition diagram is shown in Fig. 3.1. [4]
(2 marks for correct reachable set and 2 marks for correct transition diagram).

f) The transition diagram is a strongly connected graph. Hence the marked network (N,M_0) is reversible. [2]

4. Exercise

a) Denoting by $\pi_i(t)$ the probability of being in state i at time t the corresponding Kolmogorov equations can be derived as below:

$$\begin{aligned}\dot{\pi}_1(t) &= -\lambda_1\pi_1(t)+\lambda_2\pi_2(t)\\ \dot{\pi}_2(t) &= \lambda_1\pi_1(t)-(\lambda_2+\lambda_3)\pi_2(t)\\ \dot{\pi}_3(t) &= \lambda_3\pi_2(t).\end{aligned}$$

[4]

(1 mark for correct identification of states and parameters, 3 marks for correct equations).

b) The Markov Chain is ergodic as its associated graph admits a single absorbing strongly connected component (state 3). [2]

c) To find the transfer function $G_1(s)$ we proceed as below:

$$G_1(s)=[0,\lambda_3]\begin{bmatrix} s+\lambda_1 & -\lambda_2 \\ \lambda_1 & s+\lambda_2+\lambda_3 \end{bmatrix}^{-1}\begin{bmatrix} 1 \\ 0 \end{bmatrix}=\frac{\lambda_1\lambda_3}{s^2+(\lambda_1+\lambda_2+\lambda_3)s+\lambda_1\lambda_3}$$

[3]

d) Notice that $y=\dot{\pi}_3(t)$. In particular, for the absorption time we see that

$$E[\tau_a]=\int_0^{+\infty} t\dot{\pi}_3(t)dt=-G_1'(0).$$

Hence:

$$\frac{d}{ds}G_1(s)=-\frac{(2s+\lambda_1+\lambda_2+\lambda_3)\lambda_1\lambda_3}{\left(s^2+(\lambda_1+\lambda_2+\lambda_3)s+\lambda_1\lambda_3\right)^2}.$$

Evaluating in 0 the previous expression and changing sign we see:

$$E[\tau_a]=\frac{\lambda_1+\lambda_2+\lambda_3}{\lambda_1\lambda_3}.$$

[3]

e) The transfer function of the cascaded Markov Chain is simply the product of the transfer functions of individual modules. Therefore:

$$G(s)=G_1(s)G_2(s)$$

where

$$G_2(s)=\frac{\tilde{\lambda}_1\tilde{\lambda}_3}{s^2+(\tilde{\lambda}_1+\tilde{\lambda}_2+\tilde{\lambda}_3)s+\tilde{\lambda}_1\tilde{\lambda}_3}.$$

[4]

(2 marks for correct identification of cascade and 2 marks for correct expression of $G(s)$).

f) In order to compute the expected absorption time of the full Makrov chain we see that

$$E[\tilde{\tau}_a]=-G'(0)=-[G_1'(0)G_2(0)+G_2'(0)G_1(0)]=-G_1'(0)-G_2'(0).$$

where the last expression is indeed the sum of the absorption times of the first and second Markov chains in the cascade. [4]

(2 marks for correct use of formula and 2 marks for correct derivation of result).